

True8Digit project: COPA input amplifier characterization

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1 Introduction

Two identical boards were designed by INTI, built by CTU and delivered to PTB for testing:

- Board #1
- Board #2

Both boards are based on the schematic circuit depicted in figure 1, the original schematic circuit can be seen at reference [3]. The circuit is based on the original input amplifier of the DVM HP3458A [2].

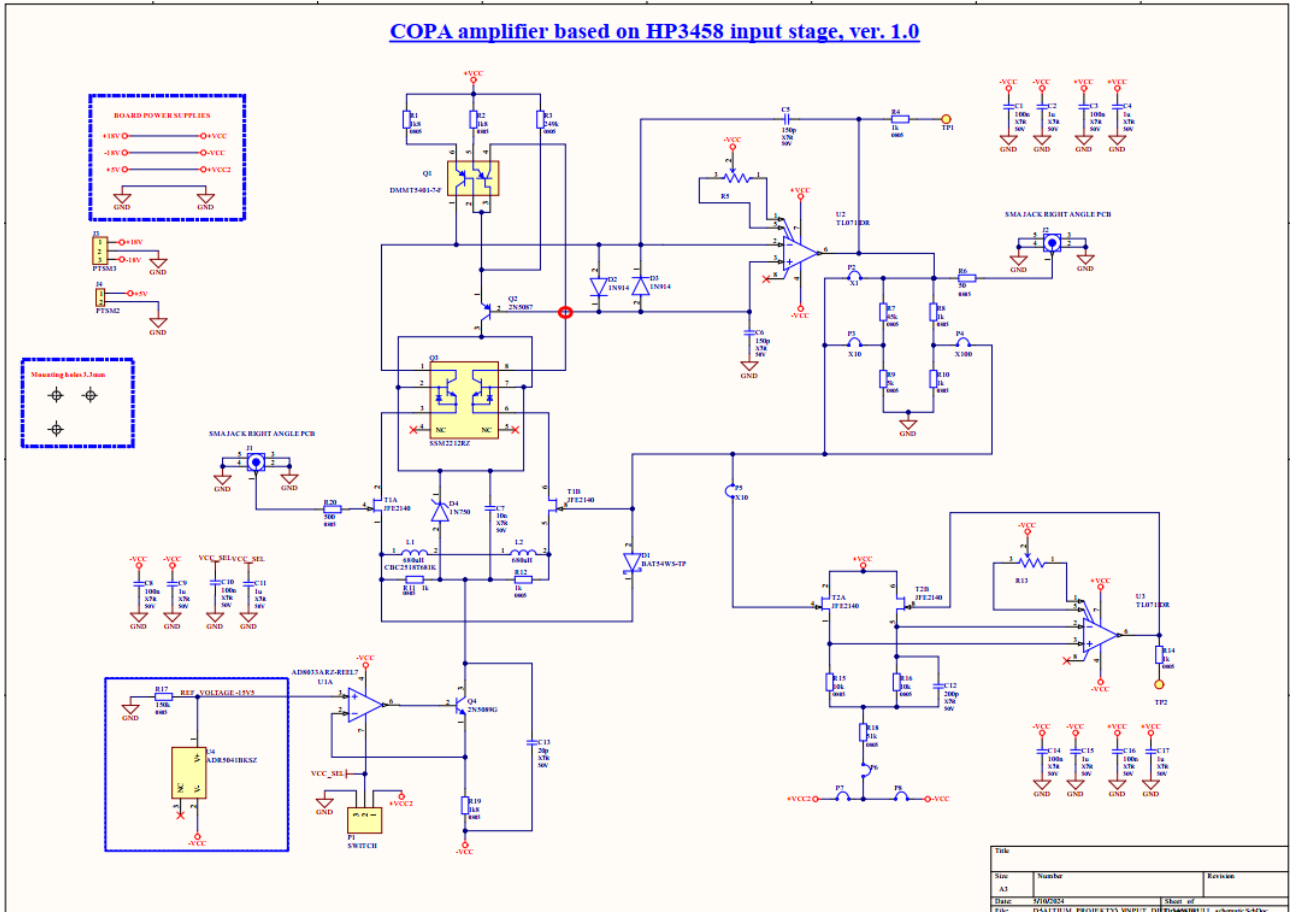


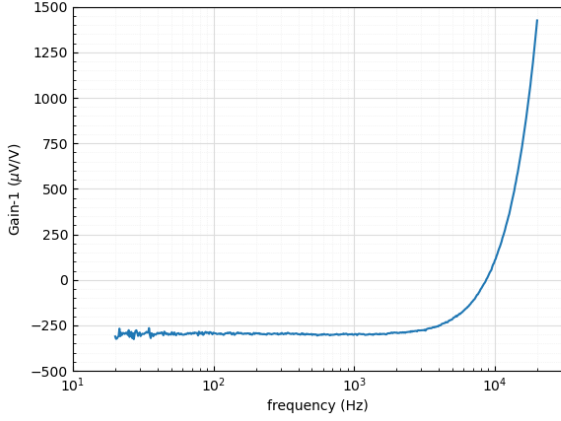
Figure 1: Schematic circuit based on LTSpice schematic ([3] and [1])

2 Test on Board #2

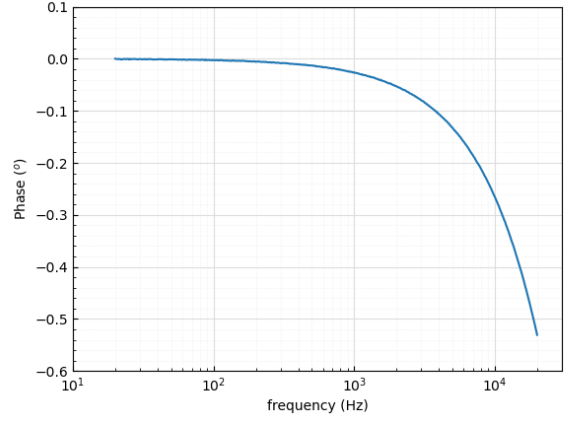
The gain, phase and total harmonic distortion (THD) as a function of input frequency were measured using the Audio Precision APx555 audio analyzer. Because on the of the digitizing channels is modified to have a high input impedance, the tests were run using channel two which has input impedance of 300 Ω , 600 Ω and 100 k Ω , therefore two sets of measures were done interchanging input and output channels. The APx555 offers the possibility of combining the peak of one acquisition with the rest of the spectrum acquired after a tuneable notch filter is adjusted to the measured frequency, thereby eliminating harmonics generated by the input stage of the digitizer.

2.1 Gain and Phase

A frequency sweep was set in order to obtain the gain and phase of the COPA amplifier, starting from 20 kHz down to 20 Hz in a linear sweep of 2000 points in total, keeping the signal amplitude fixed to the full scale value of 7.071 V to cover the full input range between ± 10 V.



(a) COPA amplifier *board #2* measured gain



(b) COPA amplifier *board #2* phase response

2.2 Total Harmonic Distortion

To measure and estimate the total harmonic distortion, the APx555 channel 2 was used to measure first the input and then the output at the frequencies listed in table 1. A full scale amplitude signal was applied to the amplifier input, that is an RMS amplitude of 7.071 V, then the THD was obtained taking into account up to the 5th harmonic.

Table 1: Harmonic Distortion results for the input and output of the amplifier including the bandwidth resulting from the sampling rate used.

Freq (Hz)	Input THD (dB)	Output THD (dB)	Bandwidth
62.5	-150.41	-143.59	DC-90 kHz
125	-147.16	-138.61	
625	-149.28	-127.87	
1000	-146.91	-123.81	
2000	-142.02	-117.43	
5000	-136.02	-108.56	
10000	-130.61	-99.00	
20000	-126.16	-88.38	
50000	-118.43	-73.15	DC-500 kHz
100000	-109.37	-46.15	

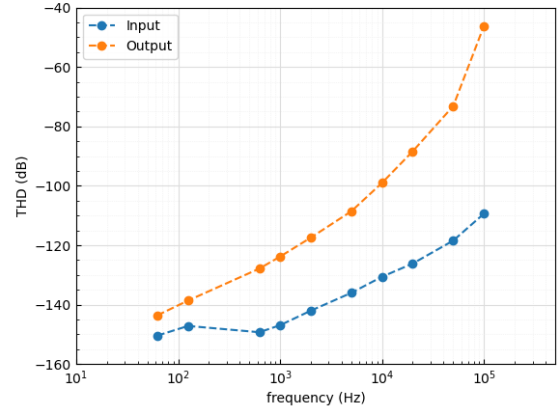


Figure 3: Total Harmonic Distortion of the circuit in figure 1 as a function of frequency

3 Tests on Board #1

Board #1 was modified as follows: the gain setting feedback resistors were removed, therefore only Gain = 1 is possible, the bootstrap circuit was disconnected from the main amplifier output because the BOOT signal is not routed on the PCB. Figure 4 depicts the circuit.

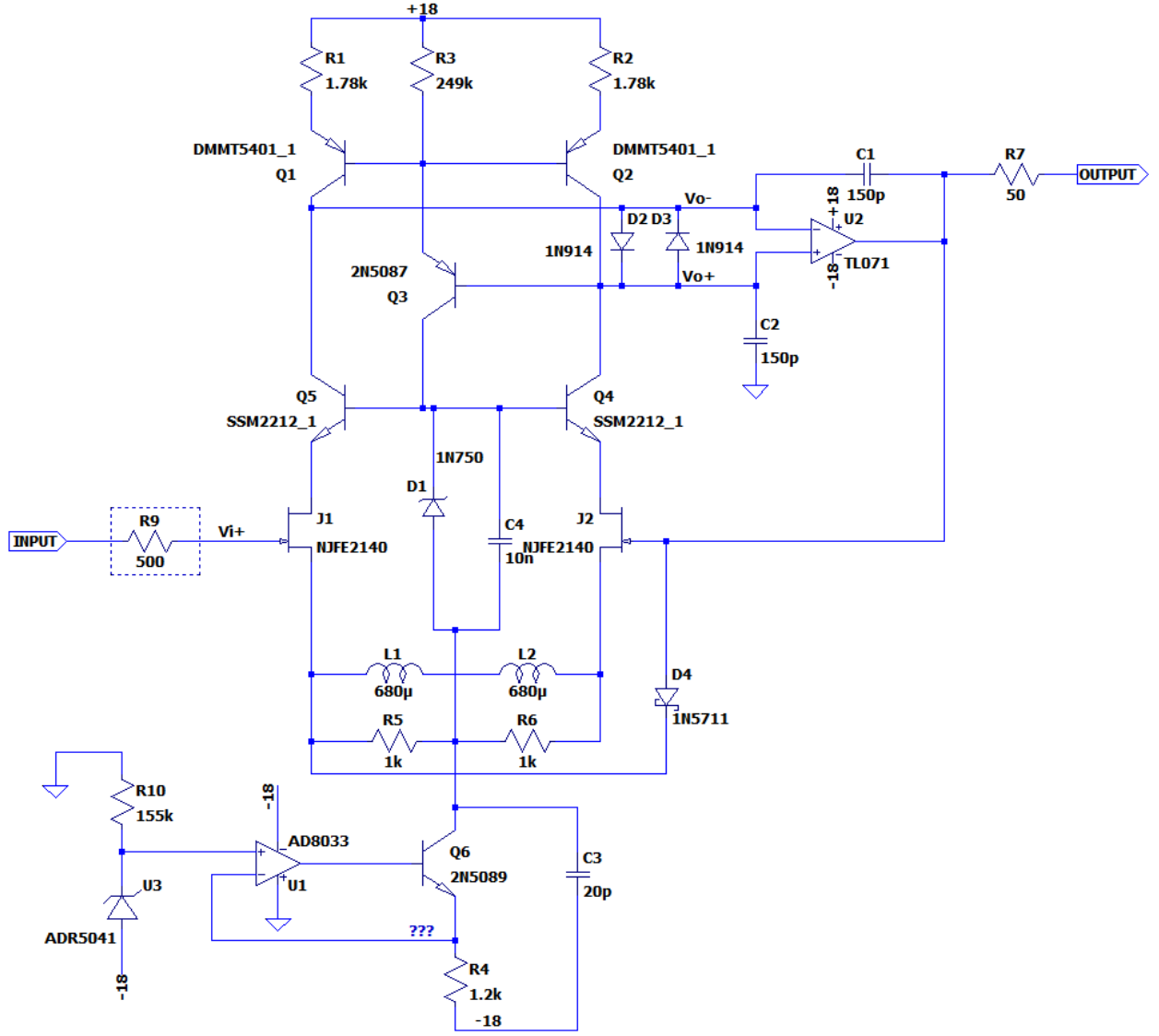
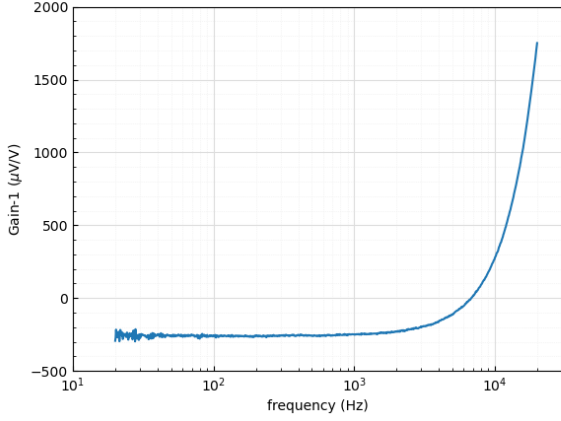


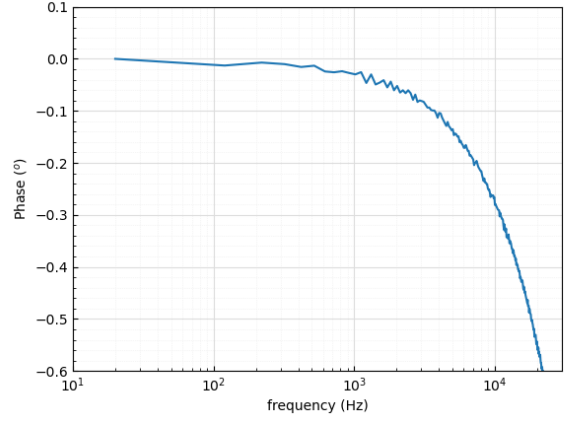
Figure 4: Schematic circuit of modified board #1

3.1 Gain and Phase

A linear frequency sweep with 2000 points was performed in order to obtain the response of the COPA amplifier, starting from 20 kHz down to 20 Hz, keeping the signal amplitude fixed to the full scale value of 7.071 V to cover the full input range between ± 10 V.



(a) COPA amplifier *board #1* measured gain



(b) COPA amplifier *board #1* phase response

3.2 Total Harmonic Distortion

The Total harmonic distortion test using up to the 5th harmonic was also performed for board #1. And was calculated using the same criteria as in section 2.2.

Table 2: Harmonic Distortion results for the input and output of the amplifier including the bandwidth resulting from the sampling rate used.

Freq (Hz)	Input THD (dB)	Output THD (dB)	Bandwidth
62.5	-151.22	-148.51	DC-90 kHz
125	-147.47	-144.36	
625	-151.36	-137.55	
1000	-147.92	-134.14	
2000	-142.79	-127.30	
5000	-135.40	-115.48	
10000	-130.60	-103.88	
20000	-125.85	-92.42	DC-500 kHz
50000	-118.87	-76.48	
100000	-109.63	-45.13	

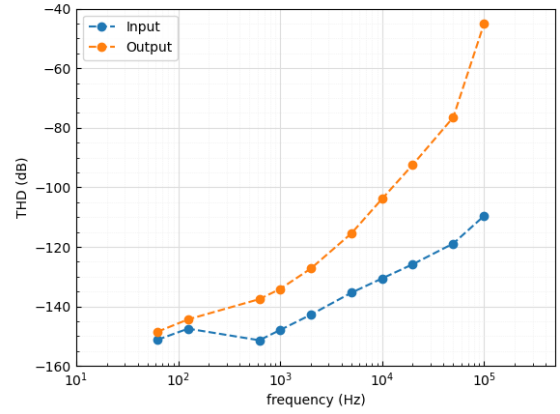


Figure 6: Total Harmonic Distortion as a function of frequency

3.2.1 Total Harmonic Distortion - Comparison

Figure 7 shows the THD of both boards, it is clear to see the improvements when removing the gain resistors and disconnecting the bootstrap circuit.

Table 3: Total Harmonic Distortion differences.
THD_1 - THD_2

freq. (Hz)	Diff. (dB)
62.5	4.92
125	5.75
625	9.68
1000	10.33
2000	9.87
5000	6.92
10000	4.88
20000	4.04
50000	3.34
100000	-1.02

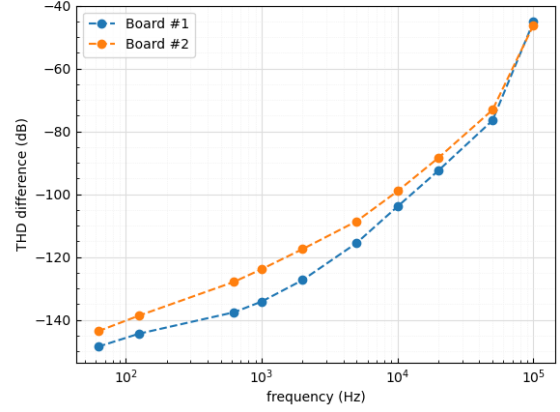


Figure 7: Board #1 and board #2 THD comparison

3.2.2 Harmonic distortion at different input signal levels

In order to characterize the harmonic distortion at different input signal frequencies and levels a test changing the input signal level while keeping the signal frequency constant was carried out as depicted in figure 8 for frequencies between 5 kHz and 100 kHz.

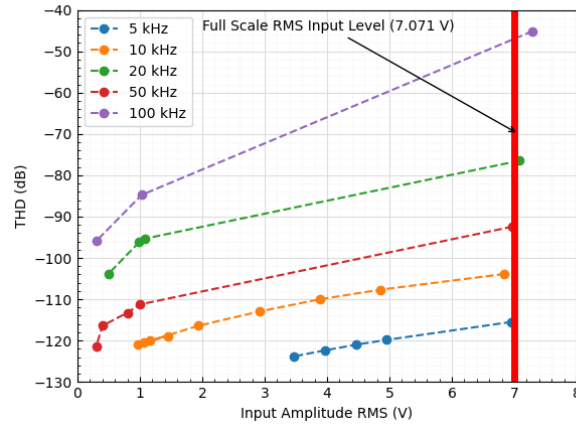
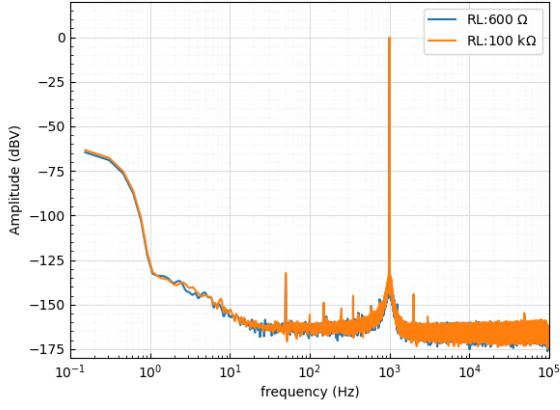


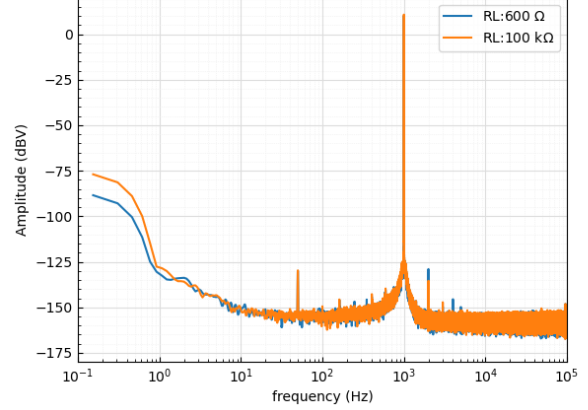
Figure 8: Total harmonic distortion vs. input signal level

3.2.3 Loading effect in the output spectrum

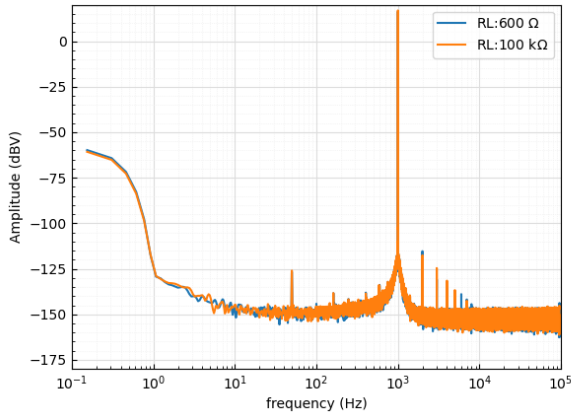
In order to evaluate the effect of loading at the output of the amplifier, the input impedance of the APx55 digitizer was changed between 600 Ω and 100 k Ω . The output spectra are depicted in figure 9



(a) Amplifier Spectrum Input Level 1 V



(b) Amplifier Spectrum Input Level 3.5 V



(c) Amplifier Spectrum Input Level 7.07 V

Figure 9: COPA Amplifer spectra at different loads values and input amplitudes

Test results are shown in table 4 where the input signal amplitude was set to 1 V RMS, in table 5 where the input amplitude was set to 3.5 V RMS and in table 6 where the input amplitude was set to 7.07 V RMS which corresponds to the amplifier full scale. The intermediate level was chosen to match the full scale value of the AD4630-24 SAR also analyzed in the project.

Table 4: Output levels at different frequencies for an input amplitude 1 V at 600 Ω and 100 k Ω loads

Harm. #	Freq (Hz)	100 k Ω		600 Ω		Delta
		Amp (dB)	Amp	Amp (dB)	Amp	
fo	1000.20	-0.12816	0.985353 V	-0.82809	0.909066 V	76 mV
2fo	2000.20	-144.15	62 nV	-149.05	35.3 nV	26.7 nV
3fo	3000.30	-156.63	14.7 nV	-159	11.2 nV	3.52 nV
4fo	4000.10	-162.09	7.86 nV	-162.08	7.87 nV	-4.48 pV

Table 5: Output levels at different frequencies for an input amplitude 3.5 V at 600 Ω and 100 k Ω loads

Harm. #	Freq (Hz)	100 k Ω		600 Ω		Delta
		Amp (dB)	Amp	Amp (dB)	Amp	
fo	1000.15	10.748	3.446672 V	10.049	3.180168 V	267 mV
2fo	2000.15	-135.39	170 nV	-128.985	355 nV	-185 nV
3fo	3000.30	-147.197	43.7 nV	-149.577	33.2 nV	10.5 nV
4fo	4000.46	-154.121	19.7 nV	-145.531	52.9 nV	-33.2 nV

Table 6: Output levels at different frequencies for an input amplitude 7.07 V at 600 Ω and 100 k Ω loads

Harm. #	Freq (Hz)	100 k Ω		600 Ω		Delta
		Amp (dB)	Amp	Amp (dB)	Amp	
fo	1000.15	16.859	6.965463 V	16.158	6.425398 V	540 mV
2fo	2000.15	-117.537	1.33 μ V	-115.307	1.72 μ V	-389 nV
3fo	3000.30	-124.607	0.588 μ V	-130.852	0.287 μ V	302 nV
4fo	4000.46	-131.498	0.266 μ V	-135.219	0.173 μ V	92.7 nV

3.3 Noise

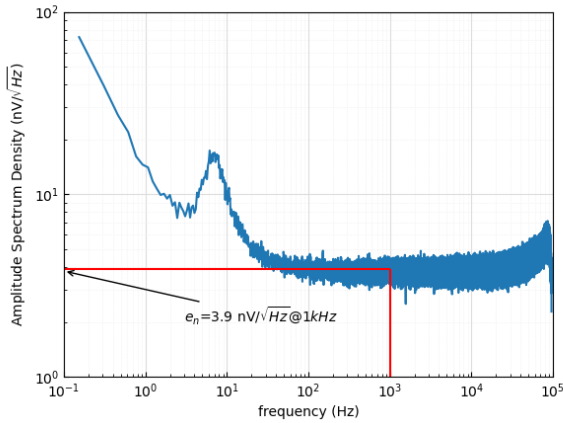
3.3.1 Equivalent input voltage noise

The input noise of the APx555 is comparable to the expected noise from the COPA, therefore PTB's PA3 was used to amplify the noise of the COPA and connected as follows: the COPA amplifier input was shorted and the output connected to a pre-amplifier (PTB PA3 Ultra Low Noise AC pre-amplifier) running both on batteries. The PTB PA3 on its internal batteries and the COPA powered by a ± 12 V battery pack. The PA3 output was connected to CH2 sampler. The AP software was configured to sample the data at 192 kHz (bandwidth DC to 90 kHz), averaging 64 windows of 1.2 M data points with rectangular windows (no window function on the software).

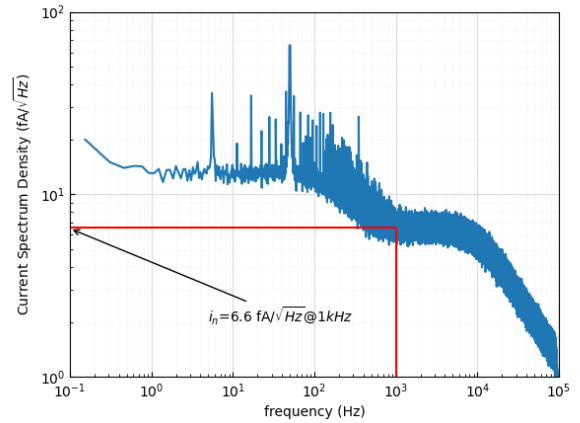
PA3 nominal gain = x60, measured gain = x59.2 and voltage noise = $0.53 \text{ nV}/\sqrt{\text{Hz}}$. PA3 voltage noise is shown in figure 11, as can be seen its noise level does not affect COPA noise measurements. Figure 10a shows the Amplitude Spectrum Density of the measured signal divided by the PA3 gain factor. The noise level at 1 kHz was determined to be $3.9 \text{ nV}/\sqrt{\text{Hz}}$.

3.3.2 Equivalent input current noise

The short circuit in the above setup was substituted with a 100 M Ω resistor. Figure 10b shows the Amplitude Spectrum Density of the measured signal divided by the PA3 gain factor and 100 M Ω resistance. The current noise level at 1 kHz was determined to be $6.6 \text{ fA}/\sqrt{\text{Hz}}$.



(a) COPA amplifier board #1 input voltage noise



(b) COPA amplifier board #1 input current noise

Figure 10: Amplifier voltage and current input reference noise

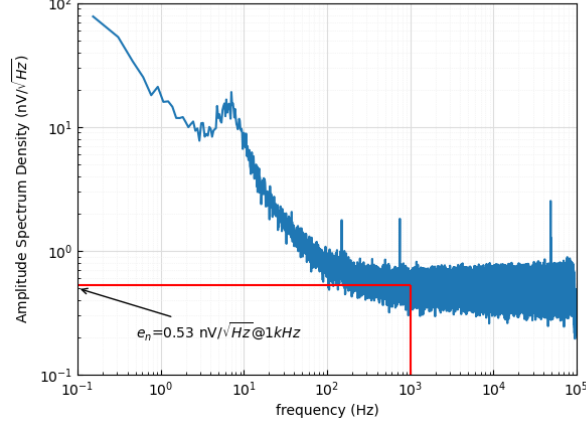


Figure 11: PA3 amplifier input voltage noise

3.4 Input Impedance

The input impedance was measured adding an external 100 M Ω in series between the output of the APx555 and the input of the COPA. and calculating the amplifier input current and voltage. The input voltage applied was 5 V RMS, then the amplifier input and output were measured by the AP on channel 2 in two steps: i) performing a frequency sweep from 20 Hz to 20 kHz (digitizer BW 90 kHz), ii) a second frequency sweep from 10 kHz up to 100 kHz (digitizer BW 250 kHz). Results are shown in figure 12 and table 7. The low frequency input impedance value results in 920.31 M Ω . Modelling the input impedance as an RC parallel circuit and performing a fitting to zero frequency an input resistance, R_i , of 1.3645 G Ω and an input capacitance, C_i of 6.384 pF were determined.

Table 7: Board #1 Input Impedance

Freq (Hz)	Z_i (M Ω)
20	920.31
1000	18.56
10000	2.22
20000	1.50
50000	1.20
100000	1.17

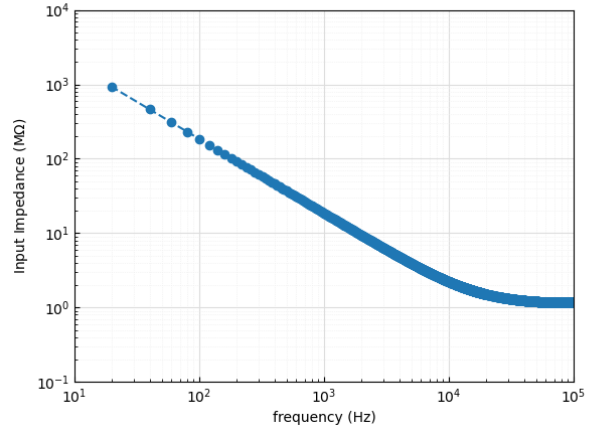


Figure 12: Board #1 input impedance in the frequency range from 20 Hz up to 100 kHz

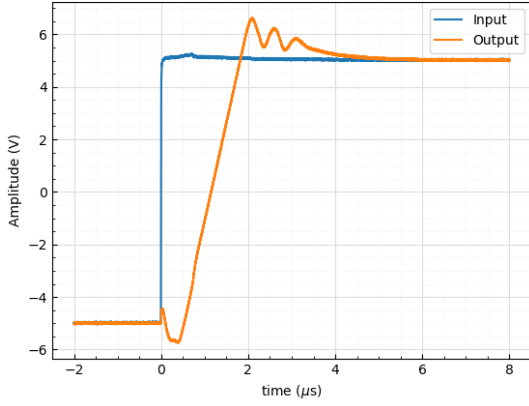
Table 8: Input RC

R_i (G Ω)	C_i (pF)
1.364	6.38

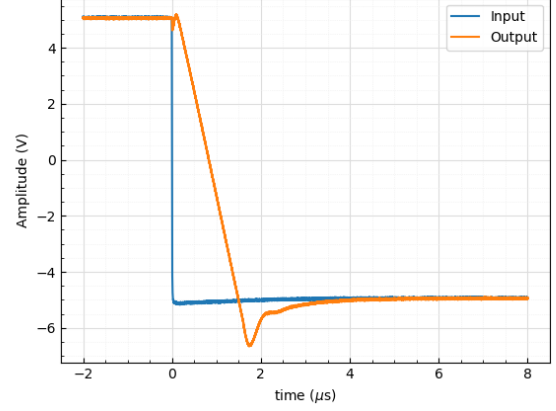
3.5 Settling time

In order to obtain the amplifier settling time, a Keithley function generator was used as source, then the amplifier input and output were captured using a high bandwidth and high resolution digital scope

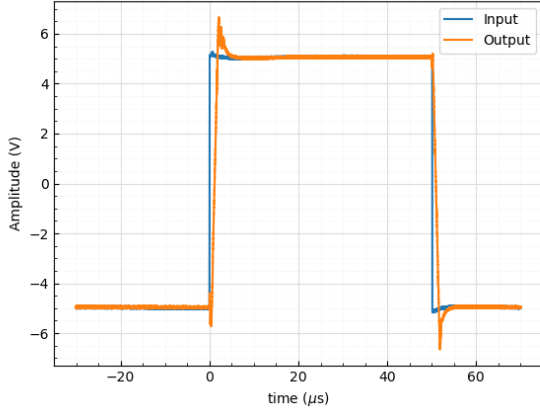
LeCroy, HDO4054, setting its digital filters to +3 bits which corresponds to a high resolution value. A squarewave signal with a peak-to-peak amplitude of 10 V and frequency of 10 kHz was used to observe and calculated the settling time. In addition, the rise-time and fall-time were calculated.



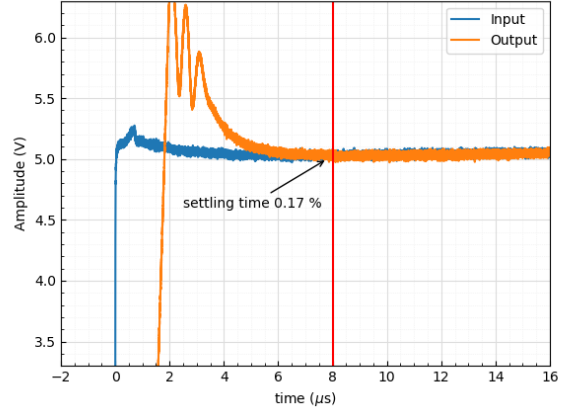
(a) Amplifier Rise time determined to be 1 μ s



(b) Amplifier Fall time determined to be 1 μ s



(c) Half period of the squarewave signal



(d) Measured settling time to 0.17 % in 8 μ s

Figure 13: Rise-time, fall-time and settling time results

The determined settling time to 0.17 % in 8 μ s can be seen in figure 13d. From figure 13a and 13b the observed rise and fall time (10 % - 90 %) are 1 μ s which are limited by the second stage of the amplifier slew rate. Therefore, one improvement could be to replace the TL071 with a high slew rate operational amplifier.

4 Modification: changing the TL071 with OPA1655

Board # 2 was modified by replacing the TL071 [5] with an OPA1655 [4] with higher slew rate. Also, on this board the gain setting resistors and the bootstrapping circuit were removed as for the board # 1. Therefore, there are two identical boards. Then, the total harmonic distortion was measured for frequencies 1 kHz, 5 kHz and 10 kHz with an amplitude RMS of 7.071 V. Results are depicted in table 9 and figure 14.

Table 9: THD of board #2 with OPA1655. Column diff stands for the THD difference against board # 1

Freq (Hz)	THD (dB)	diff. (dB)
1000	-123.547	-10.59
5000	-108.116	-7.37
10000	-99.065	-4.81

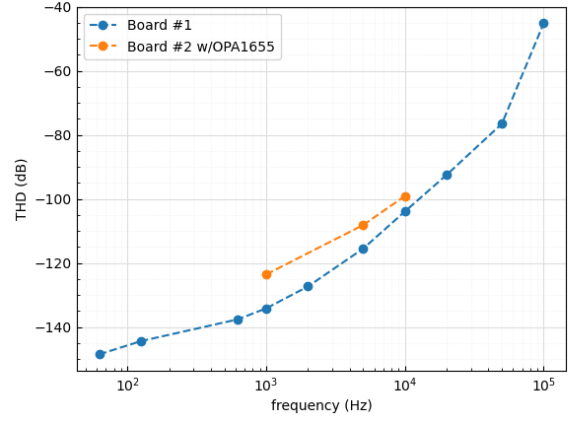
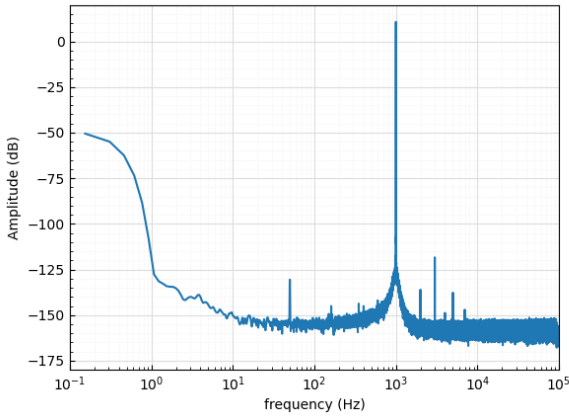
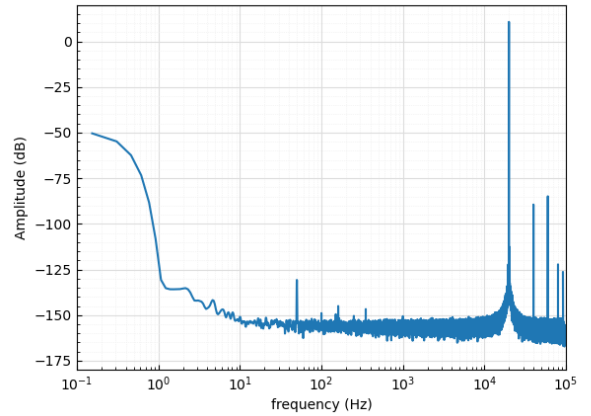


Figure 14: THD Board #1 and Board # 2 comparison

Then, to compare specifications with the amplifier datasheet its spectrum was calculated for input signals of 3.5 V RMS at 1 kHz and 20 kHz. The resulting spectra are depicted in figures 15a and 15b. The calculated THD are -128.96 dB at 1 kHz (before the change was -131 dB) and -94.27 dB at 20 kHz (before the change was -129 dB).



(a) Board # 2 spectrum 1 kHz 3.5 V RMS



(b) Board # 2 spectrum 20 kHz 3.5 V RMS

Figure 15: Output spectra of board # 2 with OPA1655 instead of TL071

5 Summary

Table 10 shows a summary of the measured features.

Table 10: Features measured results summary

Feature	Result
Gain @ 1 kHz	0.999750 V/V
THD @ 1 kHz	-134.14 dB
e_n @ 1 kHz	3.9 nV/ $\sqrt{\text{Hz}}$
i_n @ 1 kHz	6.6 fA/ $\sqrt{\text{Hz}}$
Z_i (f=20 Hz)	1.364 G Ω // 6.38 pF 920.31 M Ω
settling time	(0.17 %) (8.0 μs)
rise time (10 % - 90 %)	1.0 μs
fall time (10 % - 90 %)	1.0 μs

References

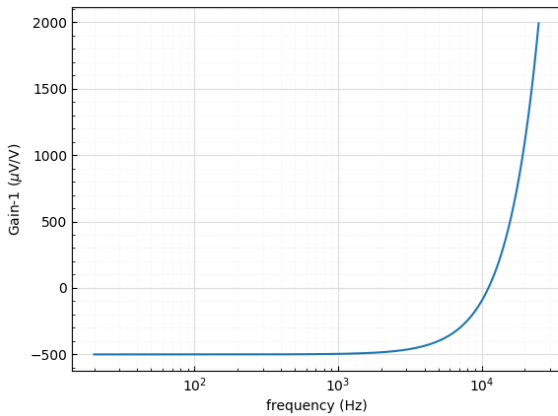
- [1] CTU. Copa pcb project. [Altium-Project-COPA-Design](#), 2024. [Online; accessed 2024].
- [2] Hewlett Packard. HP 3458 Multimeter Component Level Information Packet. Technical report, Hewlett Packard, 2001.
- [3] INTI. Copa schematic circuit. [InputDiff-3458A-FULL-RJI-03-TP](#), 2024. [Online; accessed 2024].
- [4] Texas Instrument. OPA 1655 SoundPlus ultra-low noise and distortion, Burr-Brown single audio operational amplifier. [OPA1655](#), 2022.
- [5] Texas Instrument. TL07xx Low-Noise FET-Input Operational Amplifiers. [TL071](#), 2023.

6 Appendix

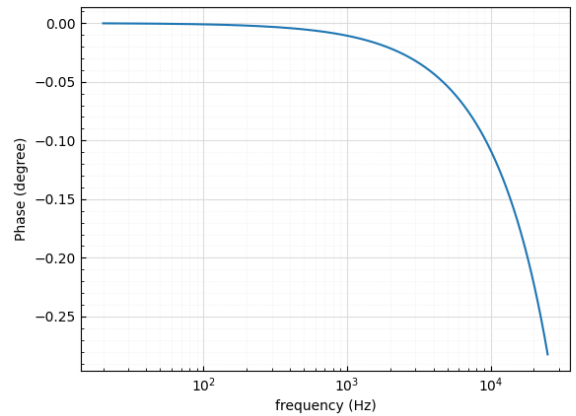
LTSpice simulation results

In this section results from simulation are shown to be compared with the measured results, more simulation can be found at the sharepoint in the document [simulation report](#).

Gain and Phase



(a) COPA amplifier simulated gain



(b) COPA amplifier simulated phase displacement

Figure 16: COPA Simulated Gain and Phase

Output Spectrum and Total harmonic distortion

N-Period=80 Fourier components of V(vo) DC component:1.21365e-005

Harmonic Number	Frequency (Hz)	Amplitude (V)	Normalized Amplitude	Phase (°)	Normalized Phase (°)
1	1000	9.995	1.000	-0.01	0.00
2	2000	3.071e-06	3.073e-07	5.80	5.80
3	3000	9.070e-07	9.074e-08	-84.95	-84.94
4	4000	3.200e-07	3.202e-08	-172.64	-172.64
5	5000	1.215e-07	1.216e-08	98.86	98.87
6	6000	4.834e-08	4.836e-09	10.39	10.40
7	7000	1.979e-08	1.980e-09	-78.29	-78.28
8	8000	8.263e-09	8.267e-10	-166.94	-166.93
9	9000	3.509e-09	3.510e-10	104.07	104.07

Total Harmonic Distortion: 0.000032%(0.000000%).

The calculated THD (dB) is equal to -129.83 dB.

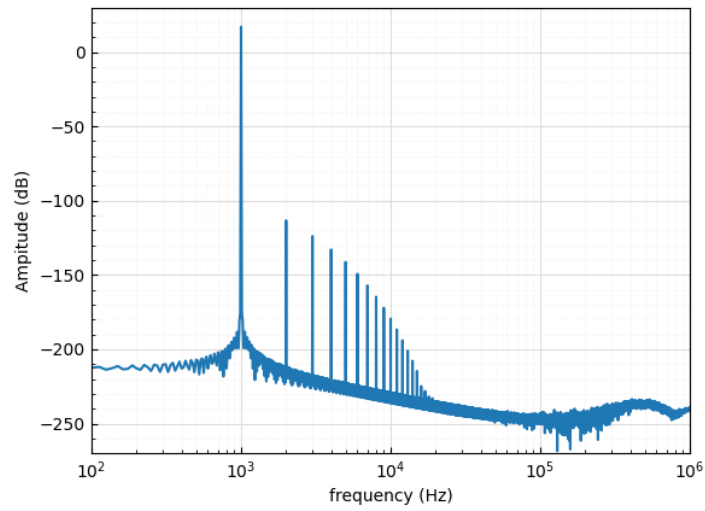


Figure 17: COPA amplifier simulated output spectrum for an input signal of 7.071 V RMS and 1 kHz frequency

Input Impedance

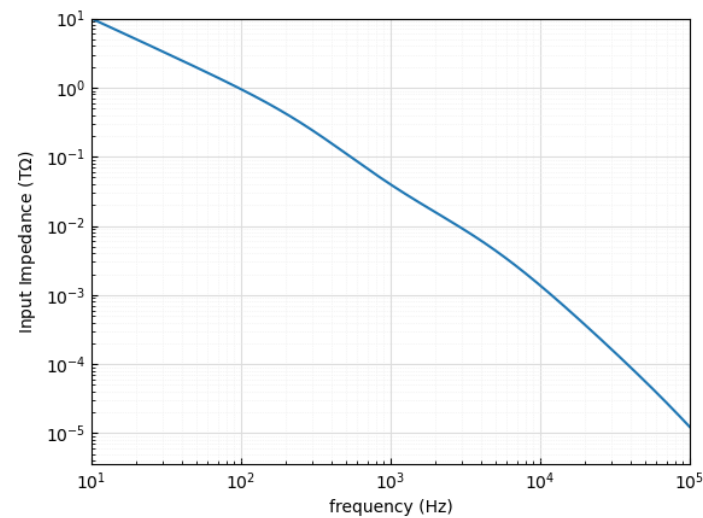


Figure 18: COPA amplifier simulated Input Impedance

Voltage Noise

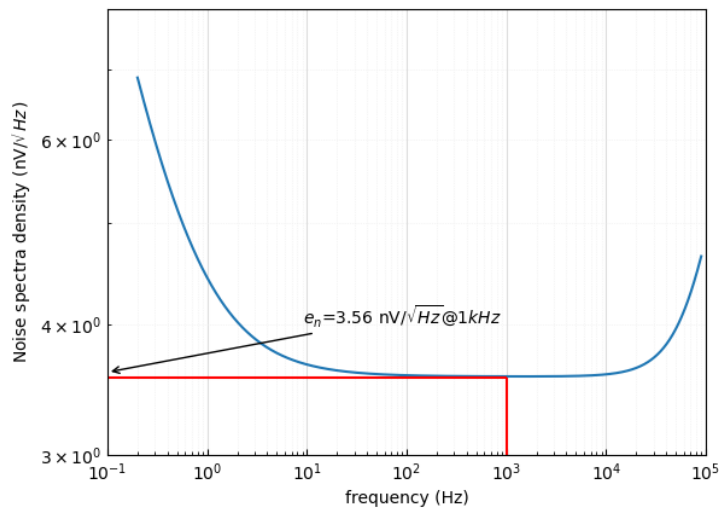


Figure 19: COPA amplifier simulated Input refered voltage Noise